Regional Quantitative Problem-Solving Circle

Module 5: Algorithmic Formulation & Invariants (Weeks 9–10)

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1 Axiomatic & Theoretical Foundation

In discrete optimization, local heuristic choices rarely yield globally optimal solutions. To prove that a greedy strategy works, we must bridge discrete mathematics with theoretical computer science by formally proving that no alternative configuration can achieve a strictly superior objective value.

Definition 1 (Greedy Algorithm & Optimal Substructure)**.** A greedy algorithm builds a solution piece by piece, always choosing the next piece that offers the most obvious and immediate benefit. A problem exhibits optimal substructure if an optimal solution to the entire problem contains the optimal solutions to its constituent subproblems.

Definition 2 (Inversion)**.** Given a sequence of elements sorted by a specific heuristic, an inversion with respect to a hypothetical optimal sequence O is a pair of elements that appear in the opposite relative order in O.

2 The Core Invariant / State Function

The formal tool to prove greedy optimality is the **Exchange Argument**.
Core Invariant Framework: If we assume a hypothetical optimal schedule O that differs from our greedy schedule G, they must differ by at least one inversion. The core invariant states that the exterior partial sums (or constraints) before and after an adjacent transposition remain strictly invariant. If we apply an exchange perturbation to O by swapping the inverted elements, and prove that the objective function does not degrade, we conclude that G is at least as good as O, making G optimal.

3 Lemma & Canonical Proof

We apply the Exchange Argument to the classic Interval Scheduling Maximization problem, proving that sorting by the earliest finish time yields the maximum number of compatible intervals.

Lemma 1 (Interval Scheduling Optimality)**.** *Given a set of n intervals, each with a start time s_i and finish time f_i, the greedy algorithm that iteratively selects the compatible interval with the earliest finish time produces a subset of maximum possible size.*

Proof. Let $G = \{g_1, g_2, \ldots, g_k\}$ be the set of disjoint intervals selected by the greedy earliest-finish-time algorithm, ordered by finish time such that $f(g_1) \leq f(g_2) \leq \cdots \leq f(g_k)$.

Let $O = \{o_1, o_2, \ldots, o_m\}$ be a hypothetical optimal solution of disjoint intervals, also ordered by finish time such that $f(o_1) \leq f(o_2) \leq \cdots \leq f(o_m)$. By definition of optimality, $m \geq k$.

We will prove by mathematical induction that for all integer indices $r \leq k$, the greedy interval finishes no later than the optimal interval: $f(g_r) \leq f(o_r)$.

Base Case ($r = 1$): The greedy algorithm explicitly picks the interval in the entire set with the absolute minimum finish time. Therefore, the finish time of g_1 must be less than or equal to the finish time of any interval in O, including o_1. Thus, $f(g_1) \leq f(o_1)$.

Inductive Step: Assume that $f(g_r) \leq f(o_r)$ holds for some $r < k$. We must show that $f(g_{r+1}) \leq f(o_{r+1})$. By the definition of a valid schedule, the interval o_{r+1} must start after o_r finishes, so

$s(o_{r+1}) \geq f(o_r)$. Using our inductive hypothesis $f(g_r) \leq f(o_r)$, we have $s(o_{r+1}) \geq f(g_r)$. This inequality proves that o_{r+1} is completely compatible with the greedy interval g_r. When the greedy algorithm selected g_{r+1}, it looked at all intervals compatible with g_r and chose the one with the earliest finish time. Since o_{r+1} was available and compatible, the greedy choice g_{r+1} must finish no later than o_{r+1}. Thus, $f(g_{r+1}) \leq f(o_{r+1})$. This completes the induction.

Conclusion via Exchange Contradiction: We have established that $f(g_k) \leq f(o_k)$. Now, assume for the sake of contradiction that the optimal schedule is strictly larger than the greedy schedule, meaning $m > k$. This implies there is at least one more interval o_{k+1} in O. For O to be valid, o_{k+1} must start after o_k finishes: $s(o_{k+1}) \geq f(o_k)$. Since $f(g_k) \leq f(o_k)$, it follows that $s(o_{k+1}) \geq f(g_k)$. This means o_{k+1} is compatible with g_k. But the greedy algorithm only terminates when there are no more compatible intervals left. If o_{k+1} existed and was compatible, the greedy algorithm would have selected it (or another interval finishing earlier), meaning G would have more than k elements. This is a contradiction. Therefore, m cannot be strictly greater than k. Since assuming $m > k$ yields a contradiction, and $m \geq k$ by definition of optimality, we must have $m = k$. The greedy algorithm is globally optimal. □

4 Seminar Heuristics & Socratic Framework

During the whiteboard defense phase, push students on the following diagnostic cues and common pitfalls:

- **Diagnostic Cue — "When to apply?":** Whenever a problem asks to minimize a maximum penalty (e.g., maximum lateness) or maximize a discrete set of resources, mandate the formulation of a sorting heuristic and an Exchange Argument.
- **Pitfall — The "Shortest Job First" Fallacy:** Students often propose a greedy strategy based on the shortest duration rather than earliest finish time. **Socratic Prompt:** *"What if we have a very short* 1*-minute task that sits exactly in the middle of a* 10*-hour block? If we pick the short task first, do we maximize total intervals?"*
- **Pitfall — Proving the Swap is Valid:** Students will swap two adjacent tasks but fail to prove the new schedule is mathematically valid. **Socratic Prompt:** *"You moved task B in front of task A. Show me the exact algebraic inequality proving that task B does not now violate the start time of the task before it."*